Correlation of Mathematical Model with AFRL RC-19 Aerothermoelastic Experiment Including an Impinging Shock

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A computational nonlinear aeroelastic solver was used in combination with the Dynamically Linearized Time-domain Approach (DLTA) Reduced Order Model (ROM) to correlate results with the AFRL RC-9 supersonic wind tunnel measurement campaign. This study is part of the Aeroelastic Prediction Workshop (AePW), High-Speed Working Group (HSWG) campaign, and it explores the 4° wedge shock impinging on the elastic clamped panel. Two different CFD solvers are used as part of the DLTA implementation, and the differences seen in each case are discussed. A detailed study on the modeling of the static pressure differential is presented, comparing the performance of an inviscid and viscous CFD solver. The effect of the static temperature differential and the mesh discretization for the turbulence modeling is briefly presented, also correlating results with the experimental data. Moreover, an additional study with a finite element model (FEM) on the modeling of the in-plane boundary condition is undertaken, and the expected experimental range for this parameter is presented.

I. Nomenclature

a, b, h	=	Plate length, width, and thickness, respectively (m)
a_∞	=	Freestream speed of sound (m/s)
d_c	=	Cavity depth (m)
M_∞	=	Mach number
$p_c(x, y, z, t)$	=	Cavity pressure (Pa)
T_∞	=	Freestream temperature (K)
U_∞	=	Freestream velocity (m/s)
$w(x, y, t)$	=	Physical displacement components (m)
$\Delta p = p(x, y, t) - p_c(t)$	=	Static pressure differential (Pa)
$\Delta T = T(x, y, t) - T_{frame}$	=	Temperature differential between the plate and its support (K)
ω	=	Frequency (rad/s)
$\psi_i(x, y, z)$	=	ith basis function for $w(x, y, t)$ expansion
ρ_∞	=	Freestream fluid density (kg/m^3)

II. Introduction

Fluid-Thermal-Structure Interaction (FTSI) in high-speed flows is a field that explores the multidisciplinary interaction among fluid forces, structural deformation, and associated thermal changers [1–8]. The aerodynamic challenges arise from the presence of strong and nonlinear phenomena, such as shock-waves, viscous and thermal boundary layers, and aerodynamic heating [1, 9, 10]. The feasibility of modeling FTSI for high-speed applications considering all these nonlinear phenomena is usually limited by the computational requirements and/or time allotted to perform these FTSI simulations. One of the benefits of working with theoretical/analytical models is that they are easy to implement and are relatively computationally inexpensive; the downside is that their intrinsic simplicity

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implies limitations in representing the complex coupling relationships among the multi-disciplinary components of FTSI. Consequently, engineers and researches usually need to choose if they will pursue a fully numerical solution that is time-consuming, with methods such as FEM, CFD, and numerical coupling, or if they can make use of mathematical/analytical methods considering simplifications to the model.

Another way of studying FTSI in high-speed flows is by performing experimental campaigns where a given phenomenon or response in the system is expected to occur. It can be in a controlled environment, such as a wind tunnel, or with a full-frame sensing model in an open environment. Either way, the goal is to capture a given set of aeroelastic responses and understand their instability mechanism during flight (or wind tunnel operation). Different experimental campaigns on the topic of supersonic and hypersonic flight have been completed [11, 12]. However, the challenge then becomes the complexity of the experiment in these high-speed conditions, as well as identifying all the relevant parameters that lead to the instability being studied. Additionally, experimental models in a high-speed context are relatively expensive and may lead to a catastrophic response. Authors such as Currao et al. [13] and Spottswood et al. [14, 15] have pursued studies to fill this gap using supersonic wind tunnel facilities, but the challenge remains. This gap is usually filled by performing companion computational studies, where researchers and engineers attempt to replicate the experimental condition using what is known in the modeling field.

In this work, results obtained in an experimental campaign by Brower et al. at AFRL [14, 16–18] are compared with computational results using a coupled aerothermal-acoustic-elastic model, as part of the Aeroelastic Prediction Workshop, High-Speed Working Group (AePW/HSWG) effort. The goal of this group is to assess the state of the art of FSTI computational models and improve the correlation with experimental results for supersonic/hypersonic flows. The measurements were performed in the Research Cell 19 (RC-19) wind tunnel at the Air Force Research Laboratory Structural Sciences Center, and the configuration comprises a thin elastic panel, coupled with a closed cavity with static fluid flow on one of its sides. The other side is part of one of the walls of a supersonic wind tunnel, where for this configuration, a 4° wedge shock wave modifies the pressure field over the elastic panel. This study will correlate these experimental results with the computational method developed by the authors, and assess the performance of the model to replicate the wind tunnel results. A similar study was previously presented by the authors [19] but targeting the no-shock configuration in the same wind tunnel measurement campaign, as well as by other authors in the field, [20, 21]. The current study focuses on the shock impingement configuration, and a Reduced-Order Model (ROM) developed by the authors [22, 23] is used to model the unsteady aerodynamics and includes the nonlinear effect of the shock wave into the aeroelastic solution.

III. Wind Tunnel Measurements: AFRL RC-19 Test Section

Figure 1 shows a schematic of the AFRL Research Cell 19 (RC-19) Mach 1.5–3 continuous supersonic wind tunnel, where measurements of the response of a thin elastic panel were performed to investigate FSI phenomena in supersonic flow. The supersonic freestream properties from the measurement are designed to produce Mach 2 at the panel section of the wind tunnel. However, measurements of the boundary layer and the cavity pressures required to excite a post-flutter response lead to the effective Mach number in the test section being $M_\infty = 1.92 \pm 0.02$ ($Re_{L_P} = 7.52 \times 10^6$), [18]. Table 1 presents the dimensions from the elastic panel, and Table 2 presents the freestream (T_0, p_0) and cavity (p_c) properties for the 4° wedge shock case. Table 2 also shows the boundary layer thicknesses at the leading edge of the panel measured in the wind tunnel ($\delta_{x=0}$); the ratio $\delta_{x=0}/a \approx 0.03$ supports the implementation of linearized aerodynamic methods such as the DLTA. Additionally, no information was provided in terms of the temperature distribution on the panel surface, leading to the initial assumption that the reported ΔT is constant over the panel surface. Further details on the wind tunnel facility, configuration, and measuring equipment are available in [14, 16].

Table 1 Aerodynamic and structural parameters (AISI 4140 Steel).

a [m]	b [m]	h [mm]	d_c [m]	E [GPa]	ρ [kg/m^3]	μ
0.254	0.127	0.635	0.051	200	7850	0.284

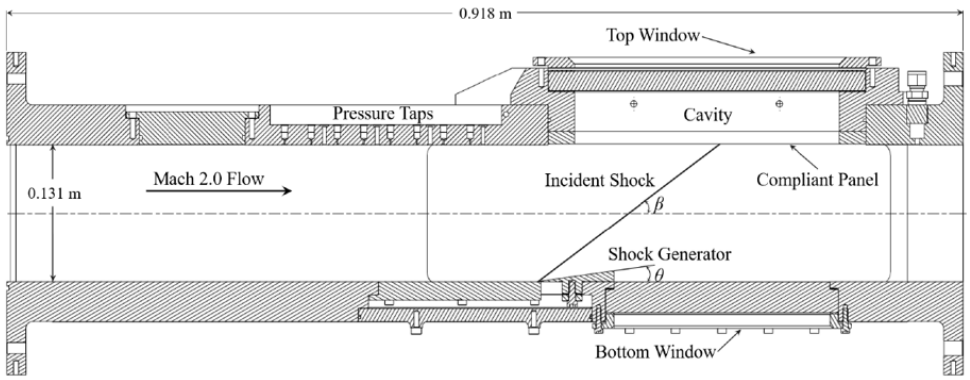


Fig. 1 RC-19 modified test section showing the pressure taps, thin panel, and cavity [14].

IV. Theoretical and Computational Model: Nonlinear Aeroelastic Implementation

The aeroelastic model is illustrated in Fig. 2, where the supersonic flow includes a shock impinging on one of the sides of the elastic panel, with the other side being a closed acoustic cavity. The mathematical-computational model includes thermal and mechanical loads acting on the panel surface, expected to have strong effects on the dynamic response of the panel, [19, 24]. Both the experimental and numerical configurations present the effect of the temperature differential, $\Delta T = T_{wall} - T_{frame}$, where T_{wall} is the temperature on the elastic panel, and T_{frame} is the temperature at the rigid frame. Moreover, there is the static pressure differential across the panel between the fluid and acoustic sides, $\Delta p = p_{aero} - p_c$. Lastly, the lateral displacement at the edges of the panel, named in-plane boundary conditions, is modeled as a set of springs, represented by a spring constant, K, as illustrated in Fig. 2. For simplicity, K is assumed constant at all edges, but a variable K can also be considered in a similar manner.

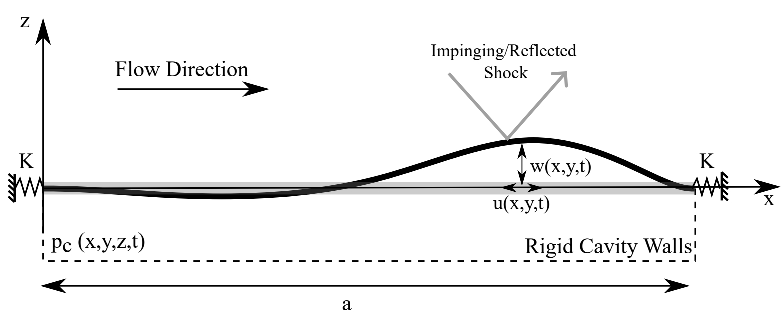


Fig. 2 Plate a) top, and b) side view with freestream flow, static pressure differential, in-plane edge stiffness K at $(x, y = 0)$, and plate temperatures.

A. Structural Model

The structural model in this study was based on the nonlinear von Karman plate equations, presented in detail with the fundamental derivations in [25, 26]. The boundary conditions of the elastic plate assume all four edges of the panel to be fully clamped with respect to transverse displacement. The mode shapes are presented in Eq. 1. Note that in Eqs. 2 and 3, the first root is zero, so $m = 2$ is the first mode shape of the clamped beam. For the y-direction, the shape function was shifted half the width of the panel, in order to have the central line the x-axis, [11].

$$\begin{aligned}\psi_m &= \cos\left[\gamma_{1_m}\left(\frac{x}{a}-\frac{1}{2}\right)\right]+\frac{\sin\left(\gamma_{1_m}/2\right)}{\sinh\left(\gamma_{1_m}/2\right)}\cosh\left[\gamma_{1_m}\left(\frac{x}{a}-\frac{1}{2}\right)\right], \quad m=2,4,6,\dots\\ \psi_m &= \sin\left[\gamma_{2_m}\left(\frac{x}{a}-\frac{1}{2}\right)\right]-\frac{\sin\left(\gamma_{2_m}/2\right)}{\sinh\left(\gamma_{2_m}/2\right)}\sinh\left[\gamma_{2_m}\left(\frac{x}{a}-\frac{1}{2}\right)\right], \quad m=3,5,7,\dots\end{aligned} \tag{1}$$

where γ_{1_m} is the $m/2$-th root of

$$\tan\left(\frac{\gamma_{1_m}}{2}\right)+\tanh\left(\frac{\gamma_{1_m}}{2}\right)=0, \tag{2}$$

and γ_{2_m} is the $(m-1)/2$-th root of

$$\tan\left(\frac{\gamma_{2_m}}{2}\right)-\tanh\left(\frac{\gamma_{2_m}}{2}\right)=0. \tag{3}$$

Similarly to previous studies on this configuration, even modes in the span direction are neglected due to the symmetry around the mid-span axis, [19, 23].

B. Aerodynamic Model

1. Unsteady Aerodynamics: Dynamic Linearized Time-domain Approach (DLTA)

The Dynamically Linearized Time-domain Approach [22, 23] is a ROM based on the assumption of dynamic linearity around a nonlinear steady state when given a small perturbation to the system. In this case, the small perturbation is modeled as a step change on the panel deformation once the nonlinear steady state is known, for example with a developed viscous boundary layer and/or a shock impinging on the panel. Since the generalized aerodynamic force is defined as the pressure field weighted by the mode shapes of the structure, Eq. 4,

$$Q_{mn}(t)=\iint\frac{\left(p_m(x,y,t)-p_c\right)}{\rho_\infty U_\infty^2}\psi_n(x,y)\,dxdy=\iint\frac{\Delta p_m(x,y,t)}{\rho_\infty U_\infty^2}\psi_n(x,y)\,dxdy \tag{4}$$

one can use the principle of input/output linear superposition to reconstruct the unsteady aerodynamics given an arbitrary motion, i.e. the coupled behavior between panel and fluid. The method is fully described in [22], and the final expression for the ROM generalized aerodynamic force is defined as

$$Q_{mn}(t)=q_m(t)A_{mn}+\dot{q}_m(t)B_{mn}+\int_0^t q_m(\tau)E_{mn}(t-\tau)d\tau \tag{5}$$

where A_{mn}, B_{mn}, and $E_{mn}(t)$ are only dependent on the Mach number and geometry shape/boundary conditions of the elastic panel. Because the configuration is a clamped panel with a small deformation amplitude, both the pressure and temperature fields are within the linear range of the panel deformation amplitude. In other words, a variation of the total pressure and temperature in the aeroelastic solution by a given factor, within the linear range of pressure and temperature for the same configuration and a fixed Mach number, means a linear variation of the matrices A_{mn}, B_{mn}, and $E_{mn}(t)$ by the same factor. The boundaries for this linear range for the pressure and temperature in the CFD solution can be obtained by assessing how much the pressure and temperature field change as the panel deformation amplitude increases. For each m structural mode required, a CFD solution is run using a known enforced motion to the structure with a small amplitude deformation (χ) to obtain an unsteady distribution, represented in Eq. 4 as $\Delta p_m(x,y,t)$.

The unsteady pressure field is obtained from a numerical CFD solution given a step change of finite magnitude. This means that the magnitude of the unsteady pressure needs to be scaled by the step change amplitude, χ, enforced as the mode shape m in the panel deformation, as presented in Eq. 6. Additionally, since the linearity is based around a nonlinear steady-state solved before the transient run, p_∞ can be substituted by the steady pressure field solution $p^{steady}(x,y)$.

$$\Delta p_m(x,y,t)=\frac{p_m^{CFD}(x,y,t)-p^{steady}(x,y)}{\chi} \tag{6}$$

For the configurations considering a viscous boundary layer and/or a shock impinging on the panel, one of the considerations is the non-uniform steady flow pressure distribution over the panel; therefore, the panel LCO can be statically loaded and deformed. Particularly for the shock configuration, an additional step is required to process the unsteady aerodynamic pressure obtained from the CFD solver, once the solution will have the steady component of the shock on the rigid panel. Because the steady pressure due to the shock impinging on the surface ($p^{steady}(x, y)$) is not affected by the enforced motion of the structure, the steady pressure distribution is removed from the unsteady Δp_m used to obtain the A_{mn}, B_{mn}, and $E_{mn}(t)$ matrices, and only the first term in Eq. 7 (pressure field scaled by χ) is used to compute the DLTA matrices. The components inside the parenthesis in Eq. 7 lead to a steady component, Q_m^{Shock}, that represents the static pressure term in the model, Eq. 8. [27]

$$\Delta p_m(x, y, t) = \frac{p_m^{CFD}(x, y, t) - p^{steady}(x, y)}{\chi} + \left(p^{steady}(x, y) - p_c\right) \tag{7}$$

$$Q_m^{Shock} = \iint \left(p^{steady}(x, y) - p_c\right) \psi_m dy dx \tag{8}$$

2. CFD Setting

The nonlinear unsteady aerodynamic solution to build the A_{mn}, B_{mn}, and $E_{mn}(t)$ matrices in the DLTA ROM was obtained using Ansys Fluent with a pressure-based solver, and using the SIMPLE scheme (Semi-Implicit Pressure-Based Segregated Algorithm), which works with the relationship between velocity and pressure corrections to enforce mass conservation and obtain the pressure field. The spatial discretization schemes are all second-order, while the transient formulation is first-order implicit. The same solver scheme has been used in previous studies implementing the same methodology using an inviscid solver, [22]. The time step size is set $\Delta t = 5e^{-6}$ sec, as discussed in [22], and the mesh setting for the RANS case involved two discretizations near the wall: $y^+ = 1$ and $y^+ = 5$. The turbulence model was the shear-stress transport (SST) $k - \omega$ within the Ansys Fluent solver. The transient CFD runs were performed with a steady result as the initial condition once the shock wave was developed over the wall and so was the viscous boundary layer in the viscous solution. The panel is considered flat and rigid during the steady solution, and the panel deformation is enforced on the flow field via mesh deformation during the transient solution.

C. Final Nonlinear Aeroelastic Model

The final coupled system of equations in modal form is shown in Eq. 9, where each set of terms is identified with its contribution to the final system. A few assumptions were made for the present study: i) The thermal components within the linear plate model are considered uniform over the panel surface. Future work will consider the heat equation in the coupled system; ii) since the panel is symmetric with respect to the x-axis, the transverse-even modes are not relevant to the final results, thus only odd-numbered modes are used; and iii) The DTLA is used instead of the Linear Piston Theory, leading the following nonlinear aeroelastic equation:

$$\underbrace{M_{mn}\ddot{q}_n + C_{mn}\dot{q}_n + G_{mn}^{(1)}q_n - \Delta T K_{mn}^{Th}q_n}_{\text{Linear plate model}} + \underbrace{D_{mnkr}^{(2)}q_n q_k q_r}_{\text{NL structural stiffness}} + \underbrace{Q_m^{Aero}}_{\text{DLTA}} + \underbrace{Q_m^{Shock}}_{\Delta p^{steady}} = 0, \tag{9a}$$

(9b)

where Q_m^{Aero} is defined as

$$Q_m^{Aero}(t) = \sum_{n=1}^{N} Q_{nm}^{Aero}(t), \text{ and } Q_{mn}^{Aero}(t) = \rho_\infty U_\infty Q_{mn}(t)$$

Table 2 presents the fluid settings for the 4° shock configuration in the experiment and in the ROM aeroelastic implementation. The same values for total pressure and temperature (p_0, T_0) were used in the CFD solution and the aeroelastic setting, and the static pressure differential was obtained from the difference between the steady flow field and the cavity pressure (p_c), and the temperature gradient was used in the ROM aeroelastic solution. Finally, the boundary layer thickness at the leading edge of the panel ($\delta_{x=0}/a$) is similar for both the experiment and computational solutions.

Table 2 **Wind Tunnel and CFD Settings, [18].**

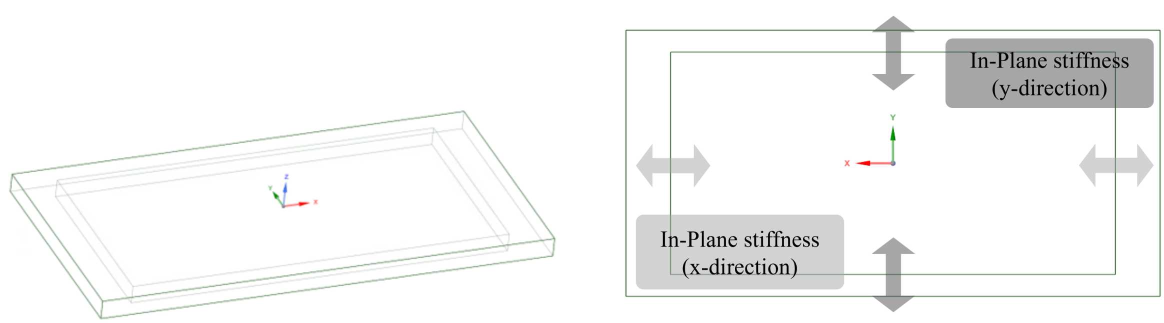
	Wind Tunnel	CFD/DLTA
T_0 [K]	345	345
p_0 [kPa]	389	389
p_c [kPa]	68.9	65, 69
ΔT [K]	13.3	15, 20
$\delta_{x=0}$ [mm]	8.6	8.0

V. Results

The nonlinear fluid-structure interaction model was implemented using the DLTA ROM to model the unsteady aerodynamic and include the nonlinear aerodynamic response into the solution. This section correlates the computational results with the response measured in the wind tunnel, and discusses implications for the sensitivity of the computational modeling.

A. In-Plane Boundary Condition: FEM *vs.* Nonlinear Implementation

One of the challenges with this configuration is the level of complexity and parametric variation during the wind tunnel operation. The main uncertainties are the static pressure and temperature differential (Δp and ΔT, respectively) since they participate in the final stiffness configuration of the flow field/panel coupling state. However, the way the boundary conditions on the panel are imposed also presents strong effects on the panel dynamic response, especially because a nonlinear structural behavior response is seen in the experimental campaign, i.e. LCO. In terms of how the clamped boundary condition at the edges was reproduced, the experimental panel was machined from a steel block, maintaining the panel edges as part of the thicker and stiffer frame from the original body. The goal was to prevent lateral displacements by the panel as the flow field acts on it. However, even with this continuous panel edge-frame connection, the aerodynamic loads acting on the panel are large enough to lead to the panel in-plane deformation near the edges, adding stress near the transition region between the elastic surface and stiff frame, and consequently modifying the in-plane boundary stiffness. The modification of the stiffness at the edges is quantified by the in-plane boundary condition factor β_{BC}, first introduced by Freydin and Dowell [28], and is defined as



(a) (b)

Fig. 3 Schematics of the model used for the FEM analysis. a) 3D view, and b) illustration of the in-plane stiffness orientations.

$$\beta_{BC} = \frac{aK}{Eh}$$

where K is the in-plane stiffness of the panel support at the edges. This parameter can be obtained experimentally by correlating the measured natural frequencies from a hammer test, for example, with the results of the computational model, where one can modify the β_{BC} value. Another option is to obtain the natural frequencies by performing a modal analysis to the panel with a preload on the structure using FEM. This was done on this configuration as illustrated by Fig. 3.

The results are presented in Fig. 4, where the first four natural frequencies are shown as the load on the panel increases (black line). The FEM results are correlated with the natural frequencies obtained considering different values for β_{BC} parameter in the aeroelastic solver with no flow field (U_∞) and no temperate gradient, also including the same values for the load (colored lines). Finally, the experimental natural frequency for the unloaded panel obtained by Brouwer et al. [16] is presented as a point at $\Delta p = 0$, for comparison.

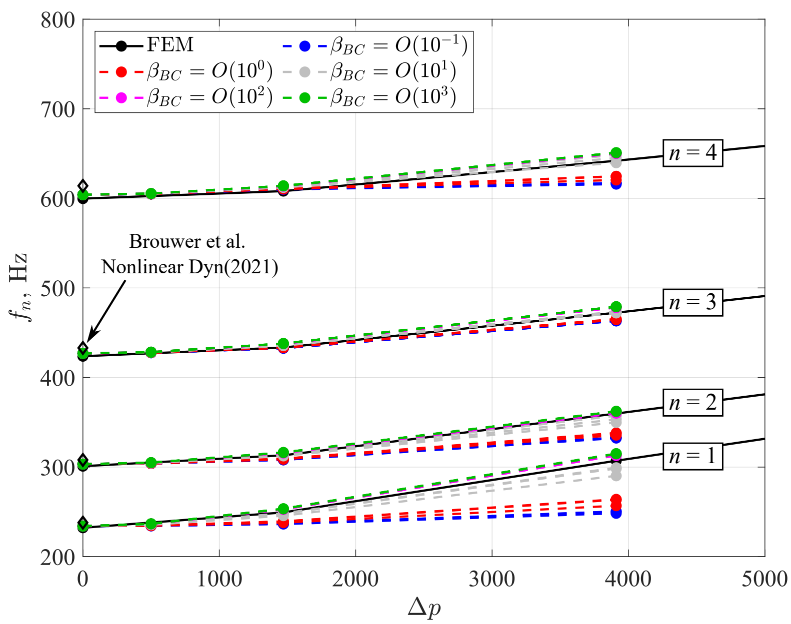


Fig. 4 Variation of the first four natural frequencies due to the static pressure differential.

As previously mentioned, the in-plane boundary stiffness at the edges, K, is assumed constant at all edges for this study, but this variable can be modeled to be proportional to the chord and/or span length (K_x and K_y, respectively. The direction of each one is illustrated in Fig. 3b). The effect of asymmetric in-plane boundary condition between the x and y direction was explored here by considering two additional cases, besides $K_x = K_y = K$: $K_x = 2K_y$ and $K_x = 0.5K_y$. All three cases are presented in Fig. 4 using the same color for a given order of β_{BC}, and one can see that the asymmetric cases did not present a substantial impact on the final values of the natural frequencies as the load increased. Moreover, the FEM results are located within the range $O(10^1) < \beta_{BC} < O(10^2)$. As the pressure differential increases, so does the added stiffness due to the added stress in the panel, enough to considerably change the dynamics of the system. More interestingly, this effect is not balanced among the modes, the first mode being the most affected by the different orders of β_{BC}. Given these results, it is expected that the correlation between the wind tunnel data and the aeroelastic solver will be more successful within the range of $O(10^1) < \beta_{BC} < O(10^2)$. However, the best way of validating this analysis is with natural frequencies obtained experimentally, considering different preload conditions on the panel, i.e. different cavity pressures.

B. Aeroelastic Response

A previous study presented some correlation between experiment and numerical solutions using a lower order model for the aerodynamics (Linear Piston Theory), while also discussing the use of the Full Potential Aerodynamics and the effects of acoustic modes to the panel dynamics, [19]. However, the same aerodynamic model is not recommended when dealing with a flow with a shock wave and its flow nonlinearity. The use of the Dynamically Linearized Time-domain Approach (DLTA) is this study is to overcome this challenge while avoiding the use of a "brute-force" coupling between FEM and CFD domains to model the RC-19 configuration with the 4° shock-wedge.

When dealing with the 4° shock wedge configuration, the nonlinear steady flow limits the implementation of the Linear Piston Theory and Full Potential Aerodynamics into the aeroelastic solver. Although the panel deformation is still within the order of the panel thickness for the 4° shock wedge configuration, the clear separation between pre- and post-shock regions leads to coupling responses that are not captured by the analytical methods. Thus, numerical solutions such as the coupling between FEM and CFD is usually used when dealing with this type of configuration. The Dynamically Linearized Time-domain Approach ROM can be used as a substitute for the "brute-force" coupling between the structural and aerodynamic, significantly speeding up the computational time and the correlation with the wind tunnel experimental data. Two different sets of results will be presented, obtained from two different CFD settings to obtain the characteristic matrices (A_{mn}, B_{mn}, and $E_{mn}(t)$): Euler and RANS. The same CFD-solving schemes were used for both inviscid and viscous solvers.

1. 4° Shock-Wedge Configuration: Inviscid Flow

As a starting point for the full RANS solution, the CFD domain for the inviscid solution was defined in a simpler form than the real wind tunnel environment. The same configuration for $\theta = 4°$ presented in Fig. 5 was considered, i.e. there are no lateral wind tunnel walls in the model. The remaining flow parameters (p_0, T_0) and settings (Δp and ΔT) were the same as in Table 2.

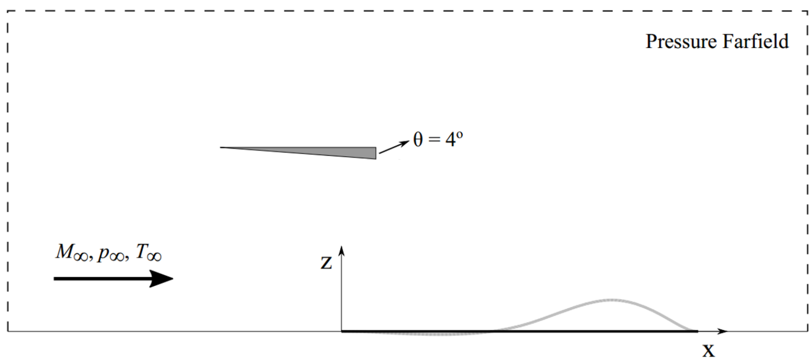


Fig. 5 Schematics for the shock wedge location in the and the flexible panel location. The shock impingement is set to be at $x_{imp}/a \approx 0.4$.

Figure 6a presents a summary of the results: the mean and standard deviation (STD) of the panel deformation, for a position at 75% of the panel length, for the last 10% of the 0.3 seconds run. For this simplified configuration of the RC-19 case, a parametric study was performed, where three different p_c and ΔT combinations are presented and compared to the mean and STD deformation of the experimental data (filled and dashed black lines, respectively). The black lines representing the experimental data were also obtained at 75% of the panel length from the mean and STD deformations provided. The experimental data is plotted as a straight line for all β_{BC} values once there is no experimental data regarding the in-plane boundary condition on the panel (only the FEM study presented in Subsection V.A). These straight lines illustrate a "boundary" characterizing how well the computational results correlate with the experimental data for each β_{BC} considered in the computational model, in terms of the mean and STD. Figure 6b presents the modal composition of the responses for the same three conditions and the final portion of the aeroelastic result compared with the experimental data obtained from [18].

While using the Euler solver for the aerodynamic composition had an intermediate role in the overall study, it provides some interesting insights into the modeling aspect of the problem. The cavity pressure and temperature gradients chosen for this analysis are comparable with the ones used in the wind tunnel measurement, Table 2, but not necessarily the same, thus seeking to explore the sensitivity of the model to these parameters. The choice for a lower cavity pressure and a higher ΔT were made to explore a scenario where the panel would not be as stiff as the conditions from the experiment and to visualize how the flutter onset and LCO would change.

In terms of the LCO amplitude and mean deformation, the comparison of the three different p_c and ΔT combinations from Fig. 6a shows how the cavity pressure pushes the mean panel deformation towards the flow side as it increases the magnitude, as expected, especially for higher values of β_{BC}. Moreover, the LCO magnitudes appear to be limited to the range of $1 < \beta_{BC} < 10$, which is one order lower than the expected range from the FEM study, but similar to the order

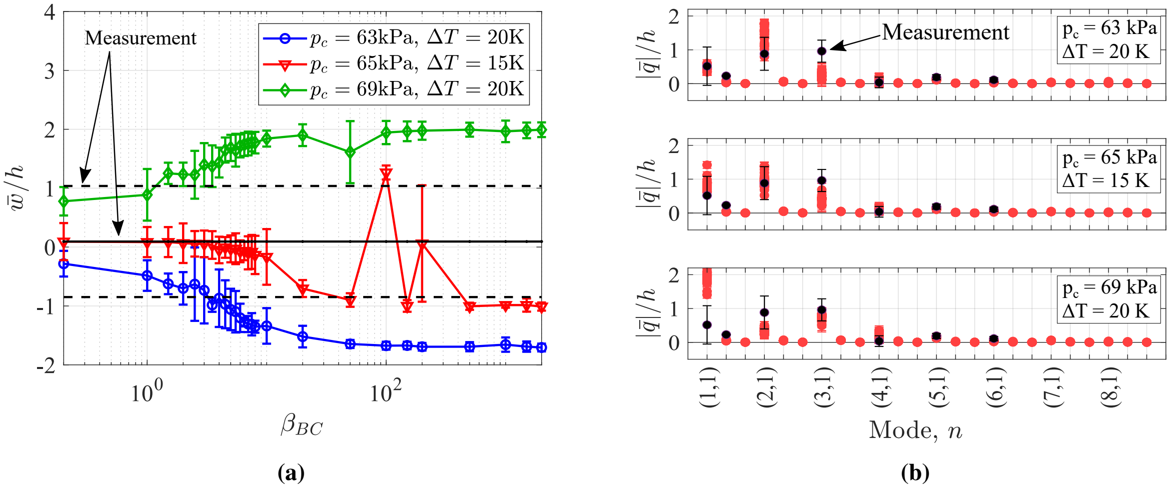


Fig. 6 Comparison with the wind tunnel data. a) Time average of the panel displacement, and b) Modal composition of the response.

seen when running the no-shock case with LPT.

Another point is that the best correlations from this set of data occur for a lower cavity pressure than the one recorded from the wind tunnel measurement, which prompts the question as to whether the issue is with the inviscid CFD solver setting or the overall aeroelastic solution implementation. Figure 7 presents the static pressure over the panel length (at mid-span) from the inviscid CFD steady solution compared with the experimental data obtained from Brouwer et al. [17]. The experimental pressure field was measured using Pressure Sensitive Paint (PSP), and an additional result was added to the figure: a RANS steady case done by Brouwer et al. When using the Euler/CFD results, both the pressure magnitude and the shock impingement location are aft of those obtained in the experiment, which explains the need for lower cavity pressure values to obtain LCO in the computational model. A further discussion will be presented later in the text in combination with the viscous results, but given the sensitivity of this configuration to the Δp parameter (and consequently $p(x, y)$), both the underprediction of the pressure field after the shock impingement, and the location of the shock-induced pressure rise being slightly downstream from the experiment explain the discrepancies seen in Fig. 6.

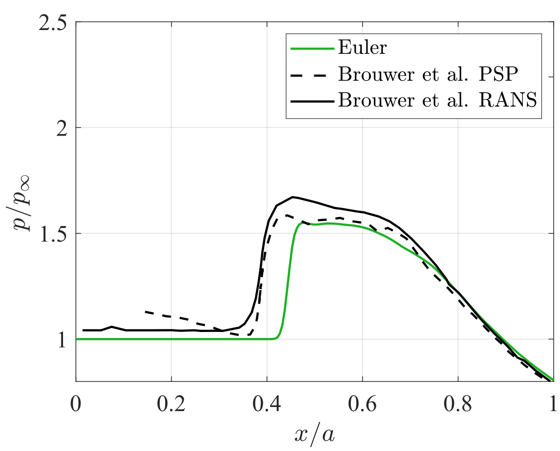


Fig. 7 Comparison of static pressure from the inviscid steady CFD solution at mid-span with the wind tunnel data.

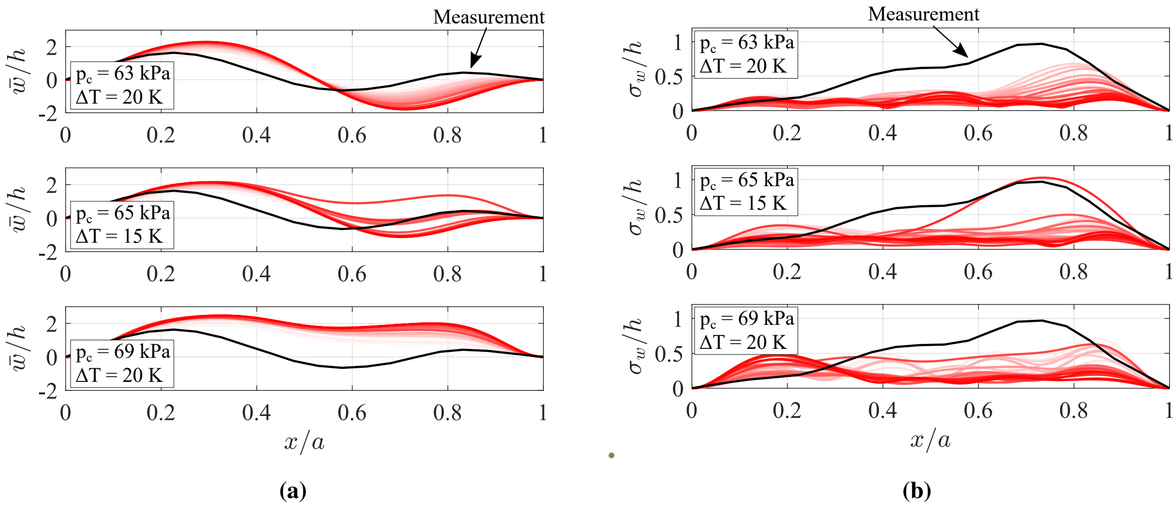


Fig. 8 Comparison of the a) mean, and b) STD of the panel deformation using DLTA and for three different flow configuration settings.

In terms of the modal composition, Fig. 6b, as the cavity pressure increases, the first and the third mode shapes increase their dominance over the response, which explains the mean and the STD profiles presented in Fig. 8. From the experimental data, it is expected that the third mode shape would be the most influential in the panel deformation. However, although the p_c = 69 kPa is the closest to the real wind tunnel condition and leads to more participation of the third mode as expected, it also leads to a static deformation that is closer to the first mode shape, increasing the first mode participation more than the third mode. Another possibility for the significant first mode participation is the larger temperature differential ΔT considered with the cavity pressure p_c = 69 kPa, which can also lead to a more significant mean deformation as the first mode shape of the panel.

Figure 8 presents the compilation of all mean and standard deviation shapes at the mid-span of the panel, over the panel length, for all values of β_{BC} considered. These results are compared with the mean and STD obtained from the wind tunnel measurement provided by the AFRL (black solid line). For all cases, the mean deformation over the panel length follows the same trend for any value of β_{BC}, except for β_{BC} = 100, p_c = 65 kPa, ΔT = 15 K. The same behavior is seen in Fig. 6a, where the mean deformation is placed towards the flow side while the rest of the results for the same condition is towards the cavity side. Similarly, for the standard deviation results, Fig. 8b, computational results for all values of β_{BC} seem to have a lower amplitude compared with the experiment, particularly around 75% of the panel length where the larger LCO magnitude is expected to occur. The exception in this case is β_{BC} = 200, p_c = 65 kPa, ΔT = 15 K case, where the magnitude is close to the experimental data.

Figures 9 and 10 present some of the mean and STD from Fig. 8 separately (the color scheme follows the same legend as in Fig. 6a). The effect of the different values of ΔT and Δp is easily seen in these plots, with a special examination of β_{BC} = 100 and 200 (in red line). It is still unclear why these values of β_{BC} present a response that is different from the other values of β_{BC}, although one of them is fairly close to what was seen in the experiment. Figure 11 presents the time response of the panel deformation at 75% of the panel length for the last 0.1 seconds for the same values of the in-plane boundary condition (the color scheme follows the same legend as in Fig. 6a). The main outcome is the chaotic nature of the results with a considerable LCO amplitude, particularly for the special case of β_{BC} = 200 at p_c = 65 kPa, ΔT = 15 K case. Up until the $t \approx 0.2$ sec, the response presented a tendency to static deformation, up to the point when a chaotic LCO started presenting amplitudes closer to what was seen in the wind tunnel. This indicates the need for longer runs to investigate if this response decays back to a static deformation or if the LCO is sustained, and if so, maintains the chaotic behavior or shifts to a periodic LCO.

Running the DLTA with an Euler CFD solver for a simple configuration of the RC-19 led to a few unanswered questions on the modeling aspect of the case, but they also provided some insights that can be further investigated using the DLTA with a RANS CFD solution. One of these issues is the inconsistency with the pressure distribution over the elastic plate when using an inviscid solver to obtain the DLTA matrices, as previously stated by Piccolo Serafim and Dowell, [23]. This can be one of the reasons why using a cavity pressure of p_c = 69 kPa led the panel to a very stiff and stable state, while a significant LCO amplitude was only seen with lower values for p_c (and consequently Δp).

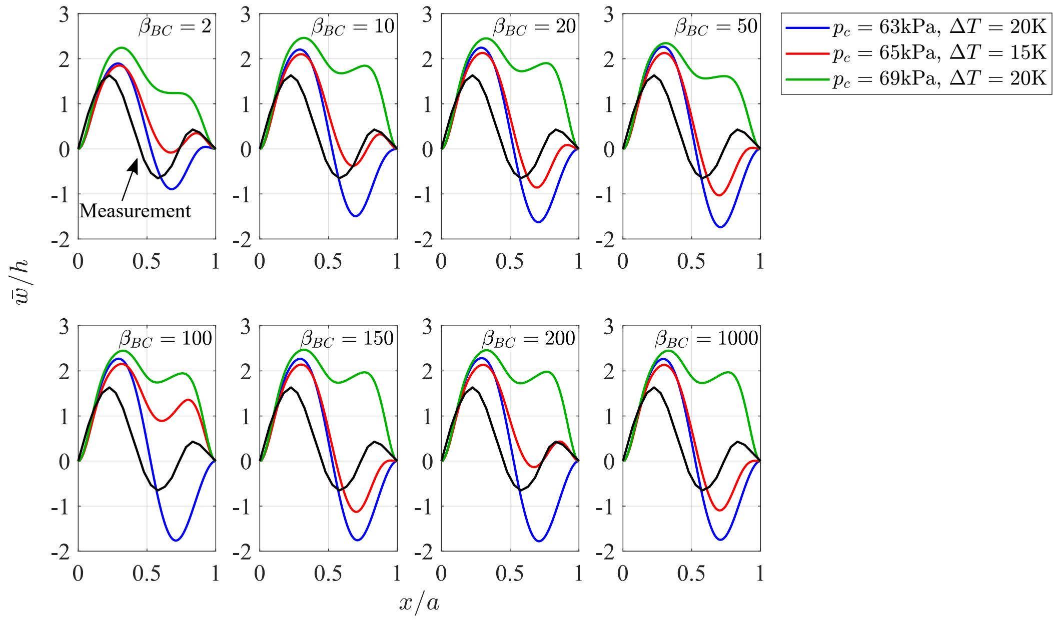


Fig. 9 Comparison of the mean panel deformation for selected values of β_{BC} using Euler/CFD.

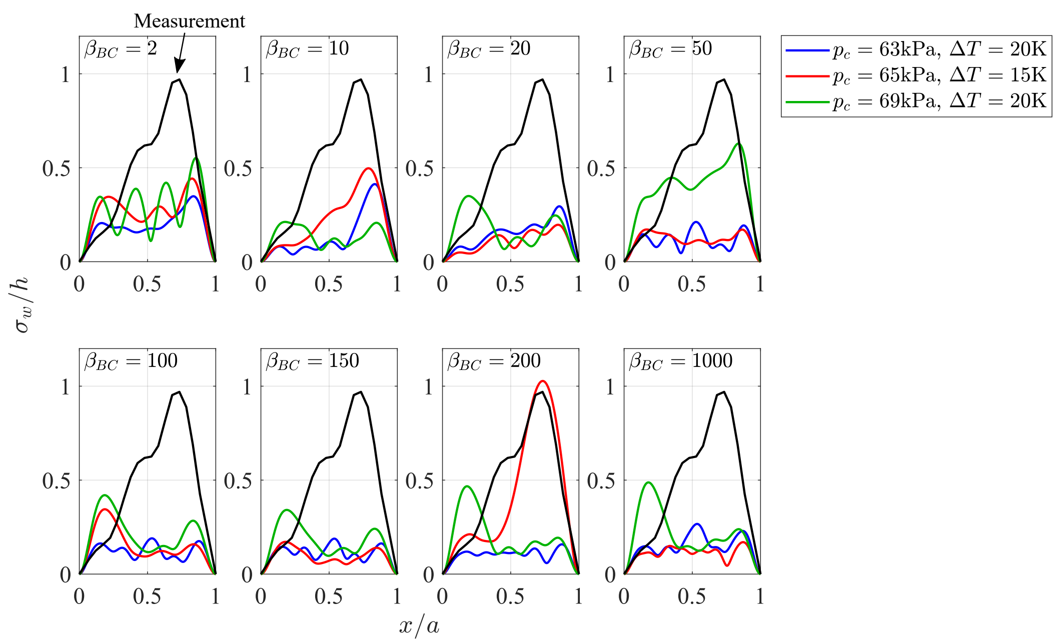


Fig. 10 Comparison of the STD of the panel deformation for selected values of β_{BC} using Euler/CFD.

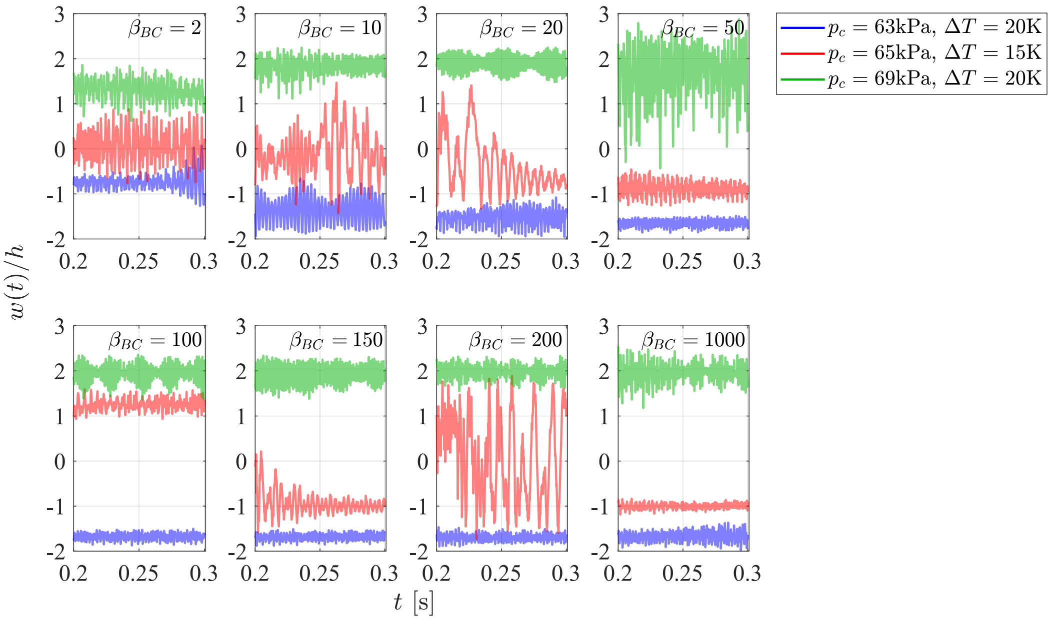


Fig. 11 Panel response in time (the last 10% of the run) for selected values of β_{BC} using Euler/CFD.

2. 4° Shock-Wedge Configuration: Viscous Flow

The CFD mesh considered for the viscous runs is presented in Fig. 12, now including the wind tunnel walls. The main difference between the Euler and the RANS CFD runs in this study is the mesh definition around the flexible panel: the viscous solution required a much finer mesh near the wall for the turbulence model. The following results consider a mesh with $y^+ = 1$ near the upper and lower walls, but Subsection V.B.3 will present results using $y^+ = 5$.

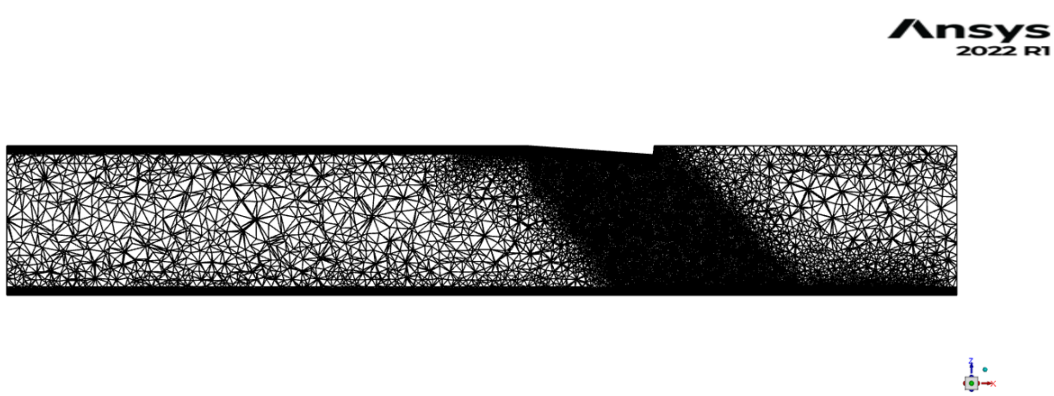
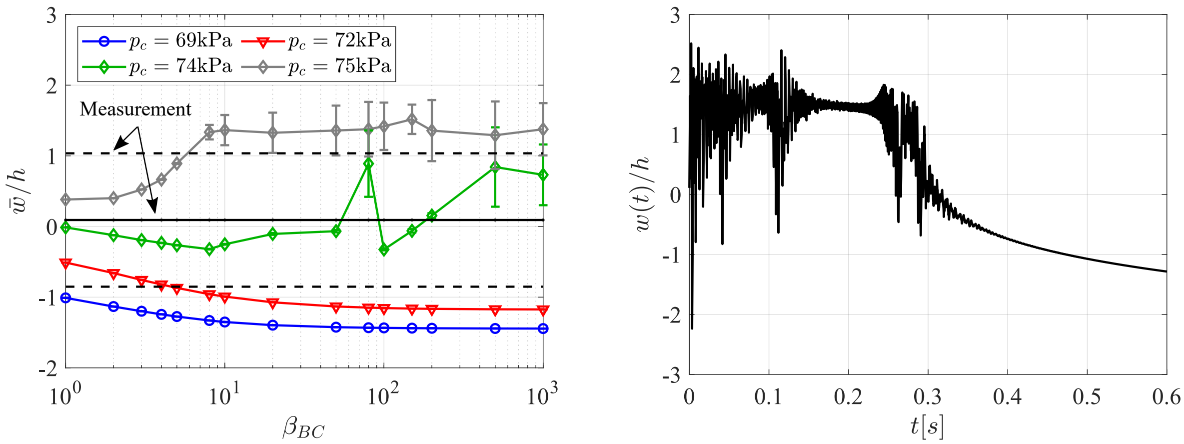


Fig. 12 Mesh setting for the 4° wedge shock configuration, for the RANS/CFD run.

Similarly to the inviscid case from Subsection V.B.1, Fig. 13a presents the mean and STD deformation of the panel at 75% of the panel length for the final 10% of a 0.3 second run, compared to the mean and STD deformation of the experimental data (filled and dashed black lines, respectively). However, the comparison is now only in terms of the static pressure differential, keeping ΔT = 15 K constant.

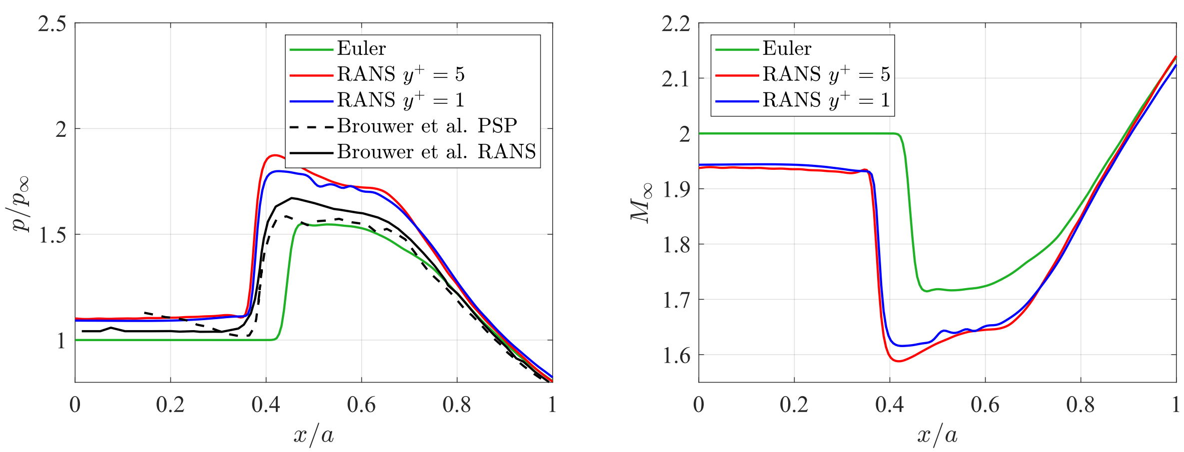


(a)

(b)

Fig. 13 a) Time average of the panel displacement (y^+ = 1, ΔT = 15K) compared with the wind tunnel data, and b) Timed panel deformation (β_{BC} = 200, p_c = 74 kPa).

One can see that the main difference of the RANS results from the Euler results is now the increase in the cavity pressure from p_c = 69 kPa to obtain LCO amplitudes at p_c = 75 kPa. As the cavity pressure increases to p_c = 72 kPa, higher values of β_{BC} start to lead to significant LCO amplitudes within the first 0.3 seconds of the run. Moreover, when running the model for longer times at the same values of β_{BC}, the LCO transitions to a static deformation with lower mean amplitudes, as presented in Fig. 13b (β_{BC} = 200, p_c = 72 kPa). LCO is seen again with p_c = 75 kPa, but since higher cavity pressure leads to a stiffer configuration, the panel can no longer deform towards the cavity as much as the experimental configuration, represented by the black lines.



(a)

(b)

Fig. 14 Comparison of aerodynamic properties from the steady CFD solution at mid-span with the wind tunnel data. a) Static pressure, and b) Local M_∞.

What is interesting from Figs. 6a and 13a is that the mean deformation of the panel obtained from the aeroelastic ROM solver is different for each CFD solver used and the experimental results, even when using the same cavity pressure. As previously mentioned, *the probable cause for this difference between the aeroelastic results using Euler and RANS is the static pressure over the panel length (at mid-span) from the CFD steady solution, as presented in Fig. 14a, compared with the experimental data obtained from Brouwer et al. [17].* While the Euler CFD result underpredicted the pressure field and the shock impingement location, the RANS/CFD computation presents a better prediction for the location of the shock impingement, but a higher magnitude for the static pressure field in the post-shock region, leading to a different Δp distribution in both cases. One can make a comparison between the effective Δp variation in this study and the wind tunnel experiments, where in the former the effective Δp variation was due to the change in the cavity pressure, [18]. Figure 15 presents the Δp distribution using the same static pressure fields from Fig. 14a for Euler and RANS ($y^+ = 1$) and using different magnitudes for the cavity pressure. Here, all pressure fields are normalized by the static pressure p_∞ at $M_\infty = 2.0$.

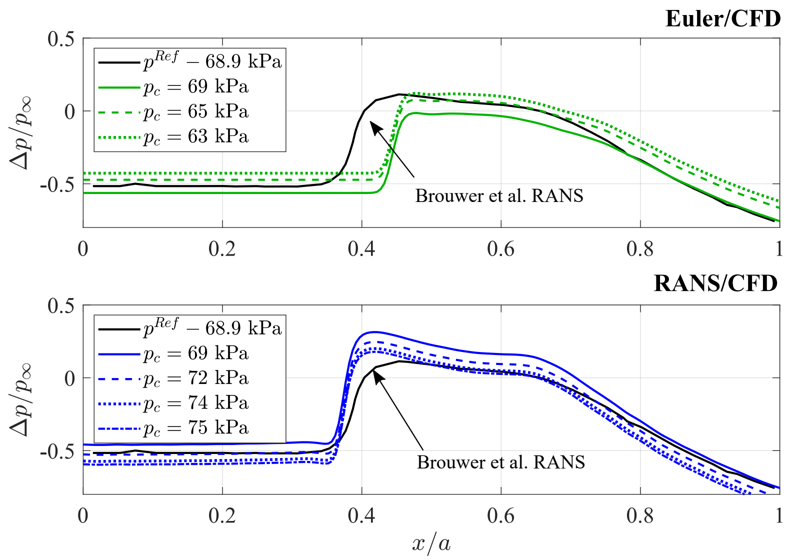


Fig. 15 Comparison of Δp obtained wuign different cavity pressures and the p_∞ distribution form Euler and RANS ($y^+ = 1$), at mid-span.

From the experimental results, changing p_c, and consequently Δp by a few kPa, the panel LCO would behave differently in terms of modal composition and amplitude deformation. In other words, the high sensitivity of this system to the static pressure differential means that even a small variation of this parameter in the simulation can lead to lower LCO amplitudes in the post-shock region if the effective Δp in the simulation is lower than that in the wind tunnel measurement. Or, if this parameter is slightly larger than the one measured in the wind tunnel, it may lead to a stiffer region on the panel after the shock impingement and consequently different LCO behavior and amplitudes. The current steady pressure field from RANS/CFD presents a larger value for p/p_∞ right after the shock location compared to the experimental data, which explains why the system requires a higher cavity pressure to lead the panel to an LCO response. Interestingly, even a small change in the cavity pressure from $p_c = 74$ kPa to $p_c = 75$ kPa, which is small enough to not indicate a significant change in Fig. 15, led to a significantly different response on the panel dynamics, as already presented in Fig. 13a, reinforcing the level of model sensitivity to the Δp parameter and its distribution over the panel surface. Unfortunately, by increasing the cavity pressure to then identify the LCO occurrence in the aft-shock region, the pre-shock area also becomes stiffer, influencing the overall panel dynamic deformation and, consequently, the flutter onset Δp value.

This discussion can also be linked to the comments presented by Piccolo Serafim and Dowell [27], where there is a balance between the two opposite trends acting on the panel when dealing with shock wave on the elastic surface (panel stiffened by the Δp effect, and a flow more likely to cause aeroelastic instability due to the smaller values of M_∞ and larger values of the dynamic pressure after the shock impingement). Once the static pressure distribution is obtained from a CFD run, the local Mach number over the panel can be computed using isentropic relations; Fig. 14b presents the comparison when using Euler and RANS, and both viscous curves have a $M_\infty \approx 1.94$ pre-shock, which is close to the values seen in the experiment ($M_\infty = 1.92 \pm 0.02$), [18].

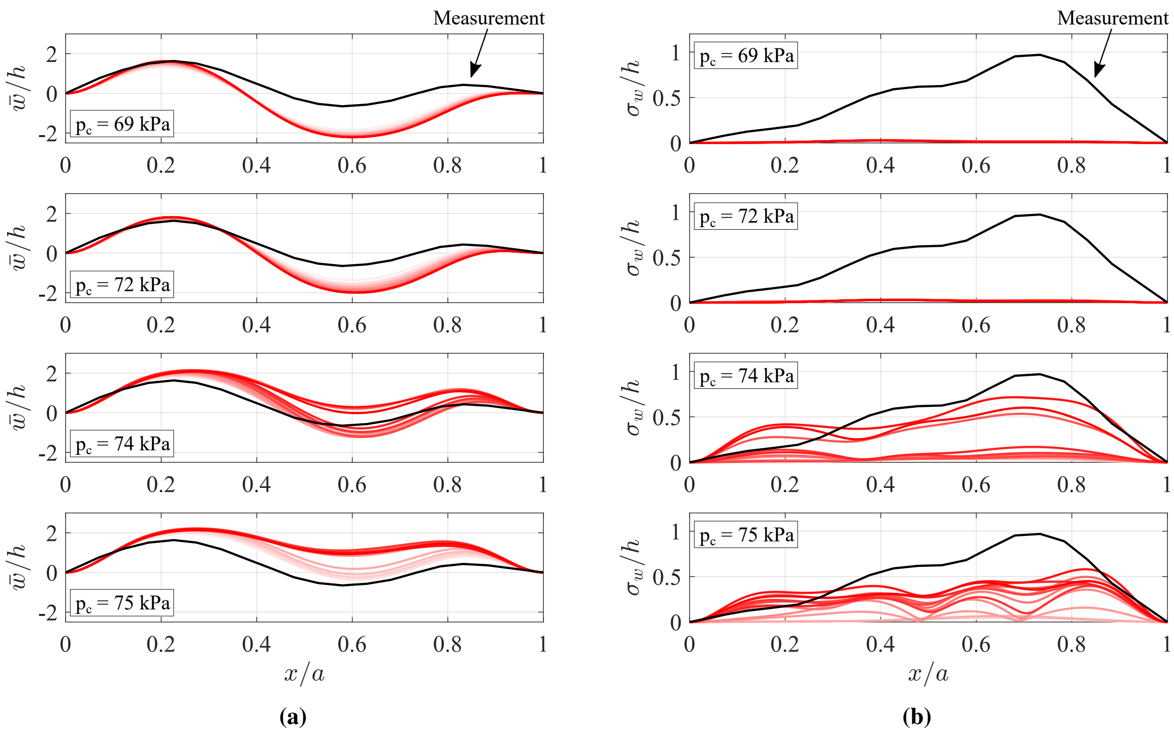


Fig. 16 Comparison of the a) mean, and b) STD of the panel deformation using DLTA and for four different cavity pressures.

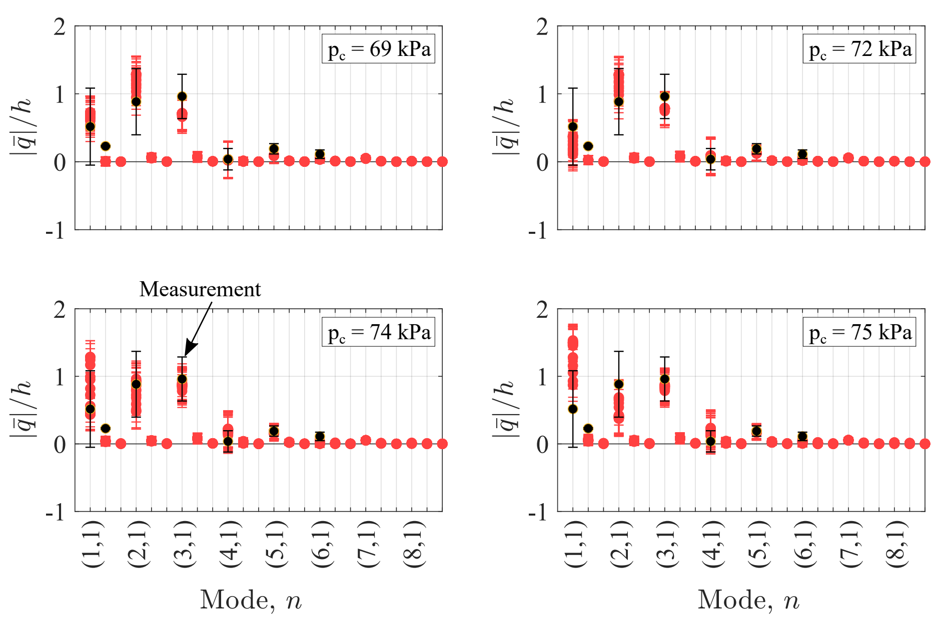


Fig. 17 Modal composition of the response for different cavity pressures.

Figure 16 presents the mean and the STD panel deformations over the panel length at mid-span. As presented in the previous figures, using a cavity pressure closer to 69 kPa leads to a very similar mean deformation before the shock impingement location, as seen in the experiment. But as p_c increases to reduce the local Δp value in the post-shock region, the deformation level of the panel before the shock also increases, affecting how the LCO shape and magnitudes will be. This relationship between the Δp, pre- and post-shock regions can help explain the modal composition of the response presented in Fig. 17, where the first mode becomes the major contribution to the panel deformation mostly due to the static pressure field pushing the panel towards the flow side of the panel.

Now looking at selected values for the β_{BC} parameter, Figs. 18 and 19 present some of the mean and STD from Fig. 13a separately (the color scheme follows the same legend as in Fig. 13a). One can see now the substantial influence of different values of the in-plane boundary condition parameter β_{BC} in restricting the edge-displacements of the panel.

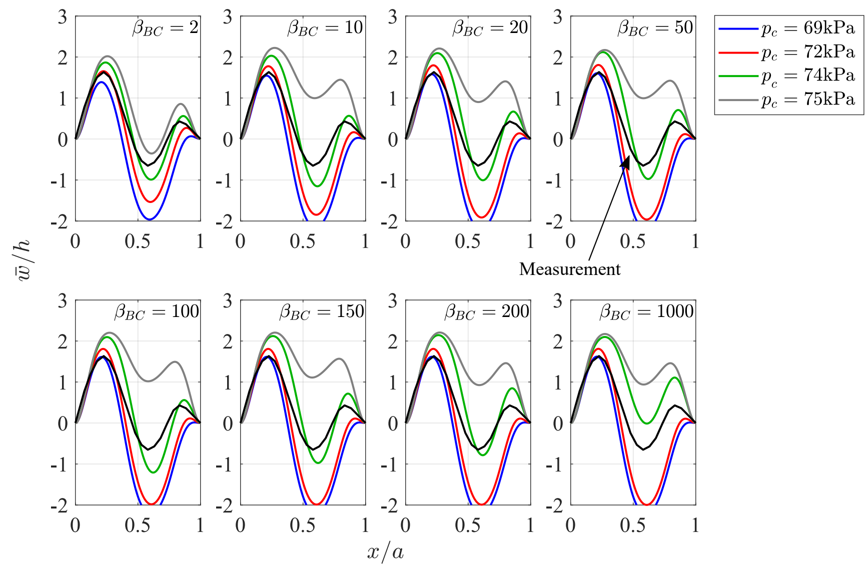


Fig. 18 Comparison of the mean panel deformation for selected values of β_{BC} using RANS/CFD.

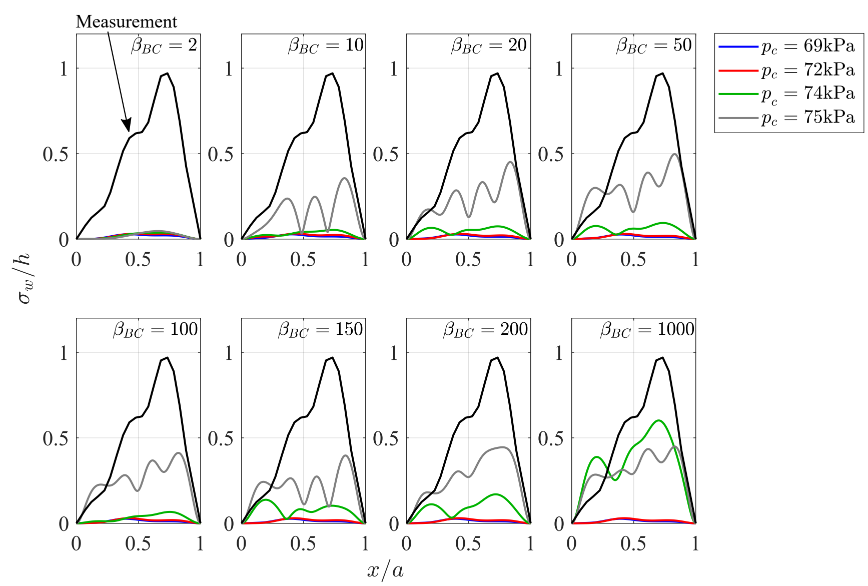


Fig. 19 Comparison of the STD of the panel deformation for selected values of β_{BC} using RANS/CFD.

Figure 20 presents the time response of the panel deformation at 75% of the panel length for the last 0.1 seconds for the same values of the in-plane boundary condition (the color scheme follows the same legend as in Fig. 13a). The main outcome is the illustration of the panel deformation decay from an LCO to a static deformation as the simulation progresses in time for p_c = 72 kPa, but a chaotic LCO is sustained for p_c = 75 kPa, at least for the simulation time range shown. See especially the results for higher values of β_{BC}. Only chaotic LCO responses were found using the computational method, which is different from the periodic LCO seen in the measured response.

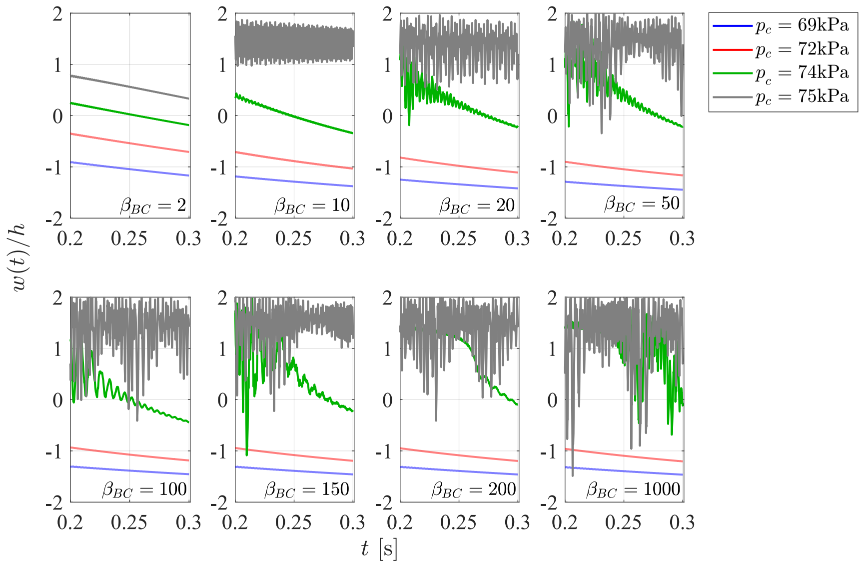


Fig. 20 Panel response in time (the last 10% of the run) for selected values of β_{BC} using RANS/CFD.

3. y^+ implications for the panel FTSI dynamics

In terms of the mesh discretization near the wall, Fig. 21 presents the mean and STD deformation of the panel at 75% of the panel length for the final 10% of a 0.3 second run, compared to the mean and STD deformation of the experimental data (filled and dashed black lines, respectively), $\Delta T = 15$ K, considering the two different $y+$ values considered so far. Two different cavity pressures are also considered, and in both cases, only static deformations are found when using the coarse mesh near the wall ($y^+ = 5$). Moreover, the mesh with $y^+ = 5$ discretization near the wall leads to an overprediction of the mean deformation of the panel.

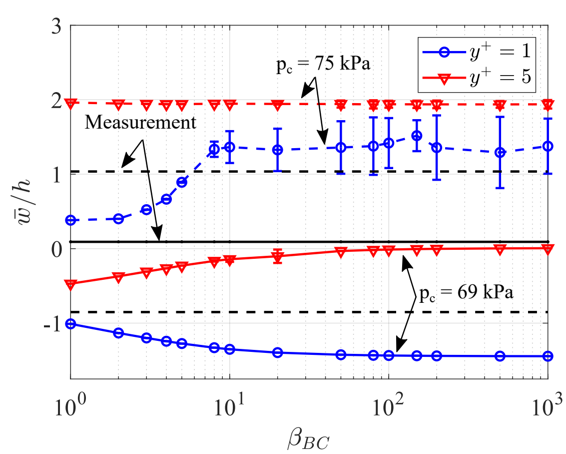


Fig. 21 Time average of the panel displacement for different mesh discretizations and cavity pressures, compared with the wind tunnel data.

Looking back at Fig. 14, there is no significant difference between the steady pressure field due to the shock impinging on the panel with $y^+ = 1$ or $y^+ = 5$, which indicates that there should not be a significant change in terms of the static pressure term in Eq. 9. Therefore, the difference seen in Fig. 21 should come from the pressure field used to compute the DLTA matrices (A_{mn}, B_{mn}, and $E_{mn}(t)$), which illustrates how sensitive these matrices are to the turbulence modeling. This result also leads to the possibility that different LCO responses could be identified in Fig. 13a, for example, if using lower values for y^+ near the flexible wall to obtain the DLTA matrices.

4. ΔT implications for the panel FTSI dynamics

The effect of the cavity pressure, and consequently the static pressure differential Δp, was briefly presented in Subsection V.B.1, where the implications of the static deformation and the LCO amplitudes were discussed. The increase in Δp leads to an increase in panel stiffness, reducing the chances for flutter/LCO to occur. The effect of the temperature gradient ΔT is now explored. In the wind tunnel experiment, two thermocouples were used: one placed on the frame and the other placed on the panel near the frame region [14], providing a measured ΔT experienced by the elastic panel. However, no information is available in terms of the mid-panel temperature during the wind tunnel operation, which leads to the question: "*Is the thermocouple on the panel at a location outside of the temperature gradient varying between the frame and the mid-panel temperature? In other words, can we assume this same value of ΔT for the entire area of the elastic panel?*"

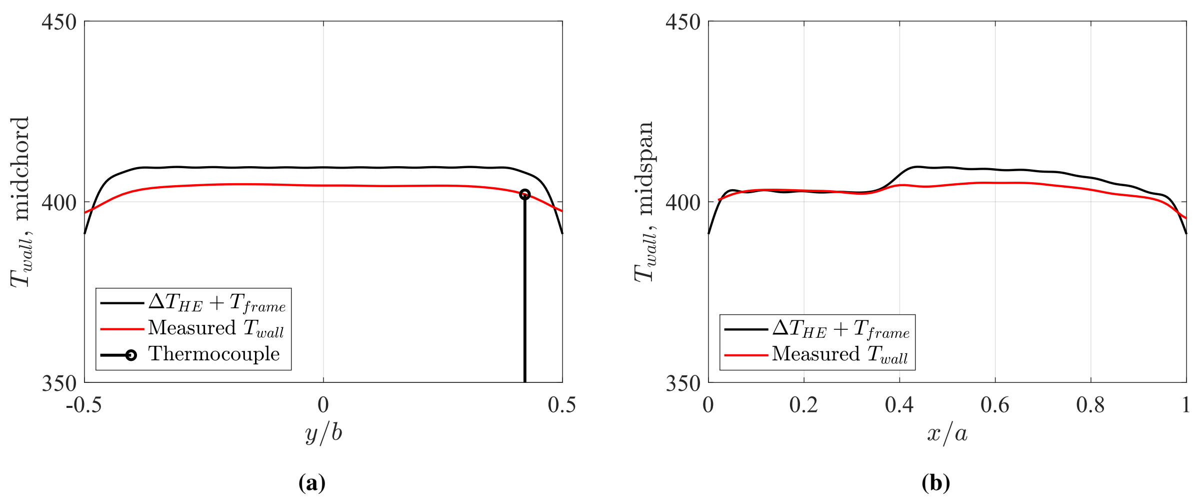


Fig. 22 Temperature distribution over the panel a) span, and b) length.

A complementary study was performed to investigate this scenario using additional data from another AFRL RC-19 measurement campaign, which was conducted under different flow field conditions (T_0 = 420K). Thus, this data is used for an independent validation of thermal models only. Still, they provide a better view of what is happening with the elastic panel when the wind tunnel is on. Figure 22 presents the wall temperature over the span and the length of the panel with the 4° wedge shock impinging on its surface, obtained using FLIR to capture a full-field surface temperature and two thermocouples. One can see that the local sensor captures the temperature in the middle of the edge region rather than the value in the middle of the panel. The figure also presents a computational result obtained from the coupling between the aeroelastic model and the heat equation in modal form, developed first by Freydin and Dowell [29], which is not explored in detail in the present study.

These results indicate that in reality, the temperature gradient over the panel is higher than what was initially indicated by the thermocouples. This has implications for the thermal buckling state of the panel as well as in the flutter onset/LCO response, as previously presented by Piccolo Serafim et al. [19]. Similar to what was done with the inviscid case, a second value for ΔT = 20 K was explored, for two cavity pressures: p_c = 69 and 75 kPa. Figure 23 presents the time response of the panel deformation at 75% of the panel length for the last 0.1 seconds for the same values of the in-plane boundary condition, y^+ = 1.

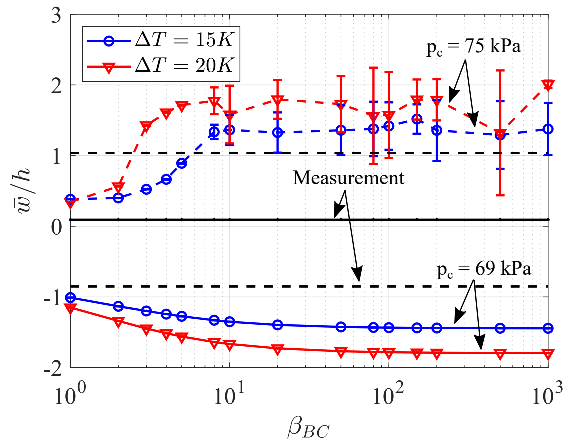


Fig. 23 Time average of the panel displacement for different ΔT and cavity pressures, compared with the wind tunnel data.

One can see that the major implication of dealing with a warmer panel is shown to be relevant only for higher values of β_{BC}. This can be associated with how much the panel is free to deform/move at the edges, i.e. a stiffer edge boundary condition is more likely to feel the thermal gradient between the elastic surface and the frame than a looser panel edge. Moreover, having a higher value for the temperature gradient between the panel wall and the frame has major implications on the mean and LCO amplitudes, but it does not seem to influence the flutter onset significantly. Figure 24 presents the mean and the STD panel deformations over the panel length at mid-span from Fig. 23, where the implications of a higher ΔT parameter are clear in the panel deformation, particularly when using p_c = 75 kPa.

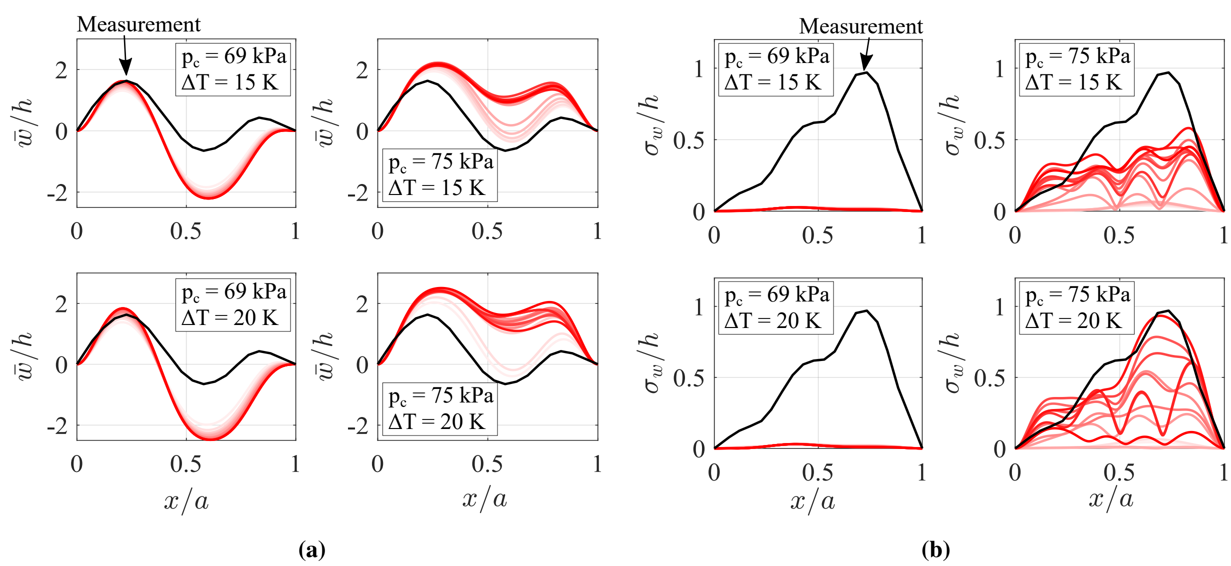


Fig. 24 Comparison of the a) mean, and b) STD of the panel deformation using DLTA and for two different cavity pressures and ΔT.

VI. Conclusion

A theoretical/computational correlation study on the AFRL RC-19 experimental case was made, as part of the Aeroelastic Prediction Workshop, High-Speed Working Group. The 4° wedge shock impingement configuration was explored and correlated with the computational model using the Dynamically Linearized Time-domain Approach

(DLTA) to model the unsteady aerodynamics considering Euler and RANS CFD solutions. The sensitivity of the model to different wind tunnel settings (ΔT, Δp, and β_{BC}) was explored and used to improve the aeroelastic modeling of the case. Particularly in terms of which CFD solver to use, it was found that there are differences in the pressure field obtained when using inviscid or viscous CFD solvers, which at first might seem to be negligible based on the fluid pressure distribution alone. However, the results presented here show that they have a significant impact in the final panel response due to their contributions to the static pressure difference Δp across the panel. Finally, the correlation between the experimental data and the aeroelastic results is promising, agreeing with the observations from experiments on the essentials of the physical phenomena e.g., buckling, flutter, and limit cycle oscillations. Overall, there is broad quantitative agreement between computations and experiments, given the sensitivity of the results to a wide range of parameters.

An assessment of the modeling of the in-plane boundary condition was made by correlating the computational results from the aeroelastic solver with a FEM model. For this analysis, there was no flow field over the panel, and the natural frequencies were obtained for different magnitudes of the static pressure over the panel. The range of in-plane stiffness for which the LCO experimental response is expected to be located is slightly different than the one previously presented in [19]. However, this range can also be validated with experimental data for natural frequencies as they change with Δp when available for this same configuration.

Future work might consider non-uniform temperature distribution over the plate with the heat equation implementation. Additionally, it is worth returning to the previous study made by the authors, [19], and exploring the use of the DLTA when modeling the no-shock configuration and investigate the sensitivity of the model to the several key parameters.

Acknowledgments

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